

Anatomy of a Deceptive Review

Comparing Linguistic Cue Profiles across Humans, a Zero-Shot LLM, and Fine-Tuned RoBERTa

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1. Introduction

Deception, the intentional misleading of others[3], is a fundamental and ancient facet of human interaction. From an evolutionary perspective, the ability to detect lies has developed primarily as a survival and self-preservation mechanism. Today, in an era dominated by mass public information and digital content, discriminating between truthful and deceptive communication has become more crucial than ever.

However, this phenomenon is marked by a persistent cognitive asymmetry: during the creation of a deception, a combination of cognitive and emotional processes are triggered[16, 14], leaving traces in language [6, 2, 13, 1]. Despite this, people, on average, are only moderately capable of distinguishing truth from lies.

This gap between "deception leaves traces" and "humans don't see them" is the theoretical core of this work.

Online reviews offer a particularly useful testbed for studying this gap. The work by Ott et al. (2011)[11] popularized this very operationalization, showing that deceptive opinion spam can be algorithmically discriminated through linguistic properties of the text, raising a question that accuracy alone cannot address.

In fact, it's unclear whether a model that discriminates well between truthful and deceptive behavior is only identifying linguistic cues related to deception or is also relying on shortcuts or specific characteristics of a given dataset. Likewise, different models might identify and consequently use different information; a supervised classifier might focus on specific patterns present in the data, while a generic LLM might instead use heuristics learned during pre-training.

The comparison could also be made with humans, whose judgments might be based on things like plausibility and personal experience, things that are difficult for models to quantify.

Different perspectives would thus emerge to solve the same problem, perspectives that might be similar or very different, but which could nonetheless begin to outline a complete profile of deception, from its deep linguistic structure to its surface manifestations, including the cognitive and perceptual strategies with which it is recognized (or not recognized) by the observer.

This is where the project's research question arises: *"What linguistic cues are reported by humans and behaviorally associated with human and machine detection of deceptive reviews, and how do these cue profiles differ?"*

The paper therefore seeks to provide a brief first step, through systematic comparison, in the discussion of the nature of lies and their detection.

2. Data and Model

2.1 Dataset

The Deceptive Opinion Spam Corpus dataset [10] was chosen for this task for several main reasons: it is quite popular, perfectly balanced, medium-sized (therefore, it works well with the limited capabilities of the free version of Google Colab) and publicly available. The original dataset contains 1,600 reviews, 800 positive and 800 negative. The project used only the positive subset, to try to mitigate confounding due to sentiment polarity. This subset consists of 400 genuine positive reviews and 400 deceptive positive reviews, all relating to 20 hotels in

Chicago. Each hotel contributes 20 reviews per class, and the data is organized into five hotel-based folds. The genuine positive reviews come from TripAdvisor, while the deceptive ones were produced by Amazon Mechanical Turk workers on request. This is important to highlight, as the difference in the collection method for the two review groups could constitute a shortcut exploitable by models: the classifier might learn to distinguish the two classes not on the basis of actual linguistic cues of deceptiveness, but on systematic artefacts related to the source (stylistic differences typical of "commissioned" writing on MTurk versus spontaneous writing on TripAdvisor, differences in average length, lexicon or formatting).

It was also checked whether the dataset used was actually present in the pretraining data of the models employed (explained in more detail in the following paragraphs), a concrete possibility, given that the dataset is public and fairly popular:

- For **RoBERTa** it can be said with good certainty that the Deceptive Opinion Spam Corpus was not explicitly used in the pretraining. The declared pretraining data are only BookCorpus, English Wikipedia, CC-News, OpenWebText, and Stories [5]. Therefore, barring an accidental copy present in one of these collections, there is no direct evidence of contamination.
- For **Qwen**, however, it cannot be ruled out with certainty. Qwen claims huge corpora drawn primarily from public web data, among other sources, but it doesn't publish a complete list of pretraining documents [15]. So Ott's corpus could theoretically have ended up in the data if a copy of it was publicly available on the web.

2.2 Qwen zero-shot

The general-purpose model used is **Qwen3-4B-Instruct-2507**. This model was chosen because the primary concern was not obtaining the most accurate prediction possible, but rather having a model that ran well on the available architecture and allowed for critical analysis of its performance. In this case, the model only saw the reviews and was not provided with other information such as gold label, hotel, fold, source, etc. The model was subjected to various tests on a 20-item pilot, during which it was found to be extremely sensitive to the prompt. The canonical prompt therefore resulted in an anonymous and A/B counterbalanced score. This solution partially mitigated the model's original sensitivity to prompts.

The log probabilities of the different options were transformed into scores (margins) that indicate how "deceptive" a review appears, and these scores were then averaged. The final result, the **deceptio_score**, is a continuous value that indicates a direction (toward true or false), not a probability. A sigmoid-transformed version, called **p_deceptive**, was also computed, but this value should not be interpreted as a truly calibrated probability (it does not accurately reflect the model's true error rate).

2.3 RoBERTa

The pretrained language model that was fine-tuned to perform a binary truthful-deceptive classification is **FacebookAI/roberta-base**, with truthful set to 0 and deceptive set to 1.

A 5-fold hotel-based cross validation was executed following the natural folds of the dataset: at each iteration, the model was trained on 4 folds (640 reviews) and tested on the unseen fold (160 reviews). A new identical model was used for each fold.

The following hyperparameters were selected:

- 3 epochs
- learning rate = 2×10^{-5}
- batch size = 8
- weight decay = 0.01

The hyperparameters were chosen in a fixed and a priori manner and allow a reasonable baseline with a standard configuration, meaning that the chosen hyperparameters are consistent with adapting the model to a dataset of this size and making reasonable use of the available computational power.

The out-of-fold predictions are concatenated into a single file of 800 reviews. This protocol preserves a more rigorous separation by hotel than a random split, although it does not eliminate the bias that arises from how the initial data was collected.

2.4 Human experiment

For the human experiment, 120 reviews were selected, regardless of model performance. Twenty-four reviews were chosen per fold, 12 truthful and 12 deceptive, for a total of 60 true and 60 deceptive reviews. To avoid overwhelming the participants, these reviews were divided into six questionnaires, each containing twenty reviews (10 truthful and 10 deceptive).

Thirty-eight people, recruited through snowball sampling and social networks, participated, for a total of 760 judgments. The six questionnaires received varying degrees of participation, but all had at least five participants.

For each questionnaire, participants were asked to read 20 reviews and, for each, choose whether they considered it truthful or deceptive and provide a confidence level for their choice on a scale of 1 to 5. At the end of the twenty reviews, they were then asked to indicate the strategies they adopted: first with an open-ended answer, then by selecting two identical checklists reflecting the cues associated with truthful and deceptive reviews. These cues were selected based on the literature, both on what humans generally observe [18, 17] in these cases and on what LLMs observe [7]. (The sample questionnaire is available at the following address: <https://forms.gle/mt7V1UUtDQtjuJ3c9>.)

The questionnaire was structured so as not to provide participants with the number of truthful and deceptive reviews, in order to not influence their choices; also the open-ended answer was asked first to prevent the checklist from suggesting strategies.

It is important to note that the sample is therefore not completely representative of the general population. Inter-annotator agreement, measured by nominal Krippendorff’s alpha, was very low ($\alpha = 0.0845$), indicating strong heterogeneity in decisions.

3. Experimental Setup

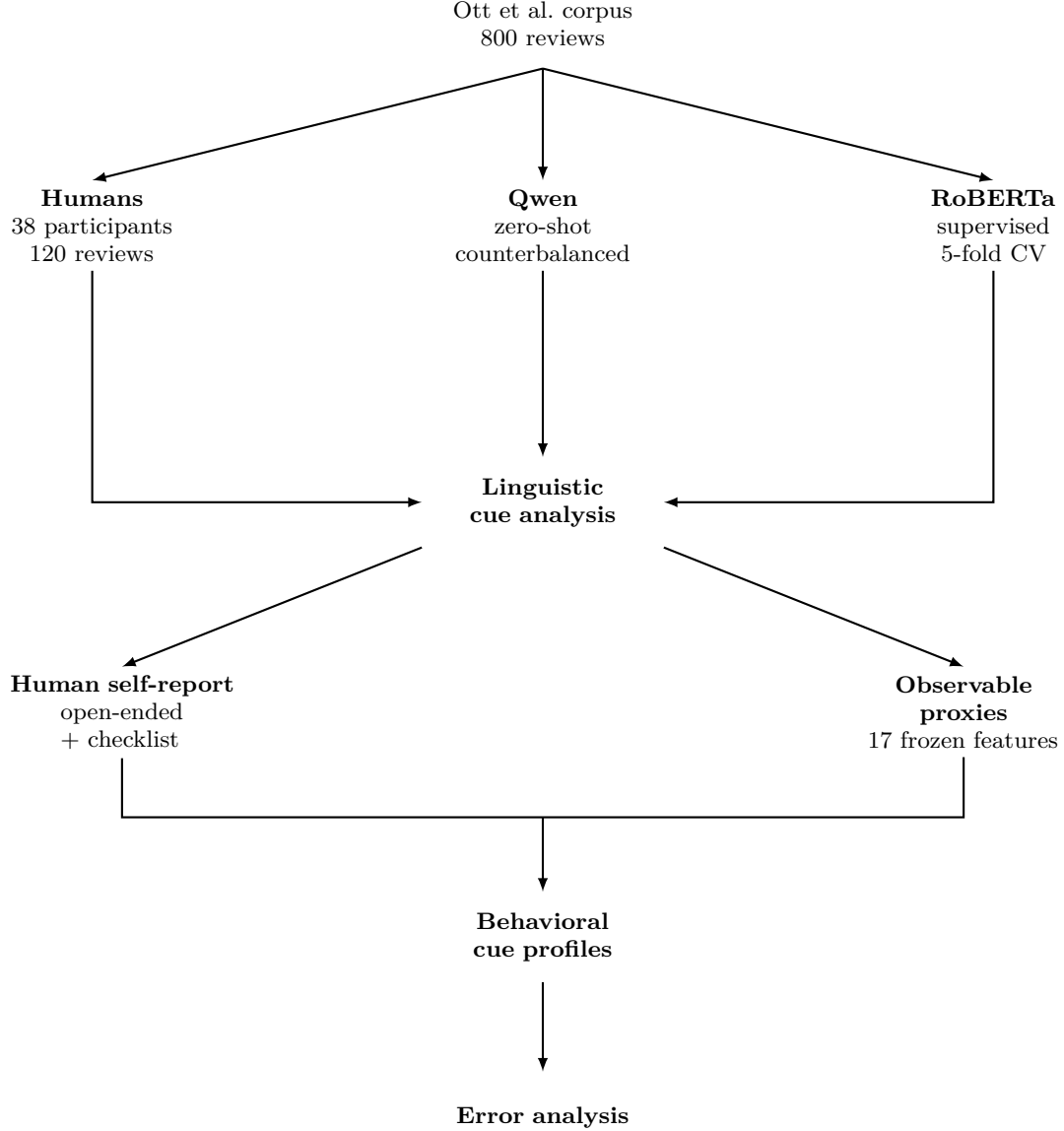


Figure 1: Overview of the experimental pipeline and behavioral cue analysis.

4. Analysis of Linguistic Cues

4.1 Cues declared by humans

The checklist included in the human questionnaire contained 12 predefined categories: *specific detail*, *situational specificity*, *mixed evaluation*, *emotional intensity*, *persuasiveness*, *length*, *grammar/spelling*, *personal language*, *promotional language*, *complexity/readability*, *repetition*, and *punctuation/emphasis*.

Among the responses obtained, one truthful answer was discarded due to its inconsistency: the participant had selected both some cues and “none” when asked what strategies they had used. The participant was not eliminated from the experiment; only this single answer was excluded.

The frequency with which participants associated a given cue with truth versus deception was then compared.

The strongest human cue for TRUTHFUL reviews was found to be *mixed evaluation*: 64.9% truthful versus 5.3% deceptive (difference: +59.6 percentage points). *Situational specificity* (references to places, times, events, and concrete situations) was also associated with truthfulness: 48.6% as a truthful cue versus 13.2% as a deceptive cue (difference: +35.5 percentage points).

The strongest cue for DECEPTIVE reviews was *promotional language* (exaggerated, promotional, or flattering language): 73.7% deceptive versus 8.1% truthful (difference: -65.6 percentage points). Other cues were also perceived predominantly as deceptive, including *persuasiveness*, *complexity/readability*, and *length*. For example, *persuasiveness* was selected as a deceptive cue in 39.5% of cases and as a truthful cue in 13.5%.

A particularly interesting result concerns *specific detail*, which did not show a clear direction: “level of specific detail” was selected as a truthful cue in 45.9% of cases and as a deceptive cue in 44.7% ($\Delta \approx +1.2$ percentage points). This is important because details clearly were salient elements in the perception of reviews, but participants disagreed on what they meant: for some, a high level of detail indicated a real experience; for others, excessive detail suggested a review crafted to be convincing. It is therefore not a simple directional cue, but rather a feature that people notice and interpret depending on context.

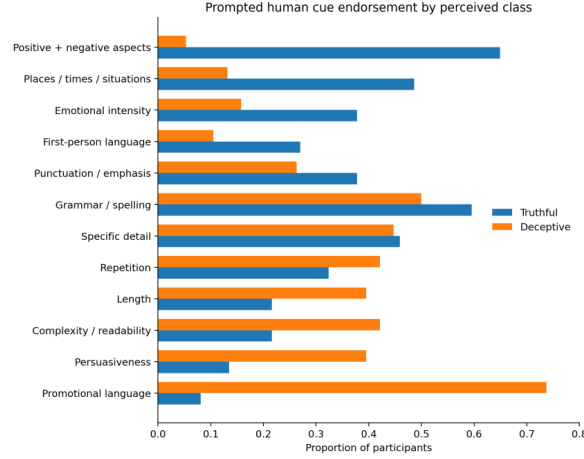


Figure 2: Self-reported cues in the checklist. The truthful and deceptive percentages have denominators of 37 and 38, respectively.

The open-ended responses were coded using a small multi-label codebook. Of the 38 participants, 10 did not provide a response, 3 explicitly wrote “None” or “nothing,” and 25 provided a substantive description of their strategy.

However, these 10 missing responses should not be interpreted as evidence that those participants did not use any strategy. Open-ended responses require considerably more effort and may therefore simply have been skipped. Furthermore, some participants may have found it difficult to verbalize the reasoning underlying their decisions, leading them not to provide a response.

Responses were manually coded, as a single response could contain more than one linguistic cue at a time. Coding was performed without knowledge of the participant’s accuracy, the gold label, Qwen’s prediction, or RoBERTa’s prediction. This was done to reduce the risk of coding responses in a way that confirmed previously known results.

In addition to the 12 predefined cues, several new strategies emerged from the responses: `VOCABULARY_LEXICAL_CHOICE`, `NATURALNESS_SPONTANEITY`, `AI_HEURISTIC`, `PLAUSIBILITY_REALISM`, and `STYLE_COHERENCE`.

Among these results, `SPECIFIC_DETAIL` once again emerged as the most frequently mentioned strategy, confirming that details are considered important, although their direction is not shared consistently across participants.

Some emerging strategies show a much sharper polarization: `AI_HEURISTIC` was unanimously labeled as *deceptive* by all participants who mentioned them. In particular, Conversely, `PLAUSIBILITY_REALISM` and `NATURALNESS_SPONTANEITY` tend to be spontaneously associated with *truthfulness*.

The most interesting of these was `AI_HEURISTIC`: some participants, despite this possibility not being suggested in any way by the questionnaire, searched for signs of AI-generated content in deceptive reviews. This is particularly relevant because it may suggest a shift in the contemporary, everyday understanding of “deception,” whereby AI-generated content may itself be perceived as deceptive.

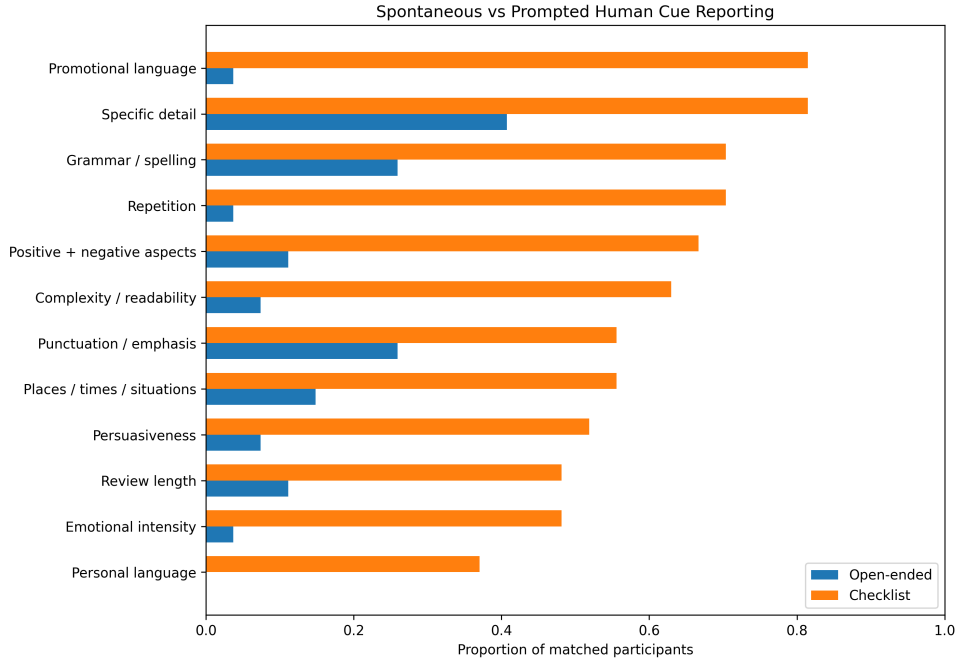


Figure 3: Checklist vs open ended responses

Comparing the cues emerging from the checklist with those emerging from the open-ended responses revealed notable differences. This does not necessarily mean that the checklist changed participants’ responses. Rather, it may indicate, on one hand, that the strategies used are difficult to verbalize spontaneously, and that the checklist subsequently helped participants articulate them; and on the other hand, that when participants were presented with a listed category, they may have found it plausible, and therefore selected it without actually having used it. This shows that spontaneous report $\neq$ prompted endorsement, which is why the two were kept separate in the analysis.

These results represent participants’ self-reported strategies and should not be interpreted as direct evidence that these cues causally determined their individual classifications.

4.2 Computational proxies

To investigate whether the linguistic dimensions discussed by participants and in previous deception research were also observable in the corpus, we defined a set of 17 transparent computational proxies.

The feature set was selected from the intersection of two sources:

1. Cues reported by participants in the human experiment,
2. Linguistic dimensions previously investigated in research on deceptive online reviews.

Then between these features only the ones that could be operationalized in a simple, reproducible, and interpretable way on the present corpus were chosen.

Previous work motivated the inclusion of dimensions related to information quantity and complexity, readability, affect, personal language, certainty and hedging, cognitive processes, and perceptual and contextual detail [12, 8].

The resulting 17 features were: number of words, mean sentence length, Automated Readability Index (ARI), first-person pronoun rate, positive- and negative-affect rates, certainty rate, hedge rate, cognitive-process rate, perceptual rate, temporal-, spatial-, and numeric-reference rates, contrast-marker rate, promotional-marker rate, exclamation rate, and repetition rate.

The proxies are not treated as complete measures of the underlying psychological constructs but as their observable and reproducible approximation. Similarly, the dictionary-based categories used here are not intended to reproduce proprietary resources such as LIWC.

4.3 Statistical analysis

Many of the selected distributions contain a large number of zeros and are highly skewed; for this reason, the two-sided Mann–Whitney U test was used. Since the dataset is of moderate size, Cliff’s delta was also computed, as it quantifies the magnitude of the difference between the truthful and deceptive groups. It was defined as follows:

- $\delta > 0$: the feature is higher in the deceptive group.
- $\delta < 0$: the feature is higher in the truthful group.

The larger $|\delta|$, the greater the separation between the two groups.

However when testing 17 features simultaneously, the multiple comparisons problem arises: the more tests are performed, the higher the probability of obtaining false positives. To address this issue, the Benjamini–Hochberg procedure was applied. Out of the 17 features, 14/17 remained statistically significant after FDR correction. This indicates that the two classes in the dataset exhibit many systematic linguistic regularities.

5. Results

5.1 Predictive performance

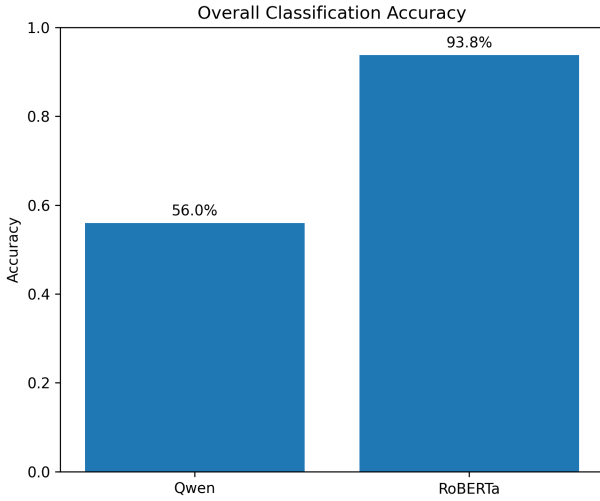


Figure 4: Qwen vs RoBERTa accuracy across all 800 reviews

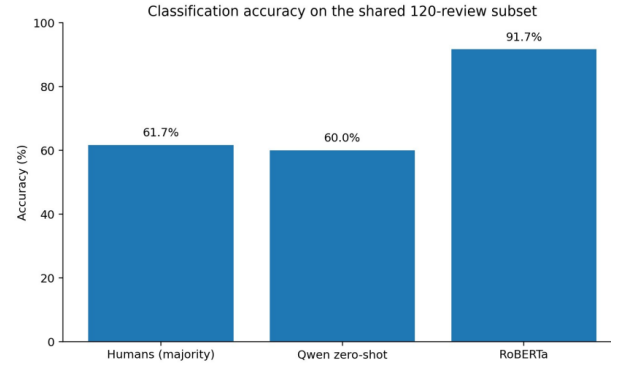


Figure 5: Accuracy on the shared subset of 120 reviews. The Humans bar uses majority voting with ties treated as incorrect.

On the full corpus, Qwen achieves an accuracy of 56.0%, compared to 93.75% for RoBERTa (out-of-fold). Qwen showed a clear tendency to over-predict the *deceptive* class, with 67.5% of reviews classified as such.

Looking only at the shared subset of 120 reviews, the human majority vote reaches 61.7% (treating the three tied cases as unresolved/incorrect), Qwen reaches 60.0%, and RoBERTa 91.7%. Human judges displayed a pronounced *truth bias*, consistent with prior findings in the literature[4, 9].

Restricting the analysis to the 117 reviews with a clear human majority vote, McNemar’s exact test finds no statistically significant difference between humans and Qwen ($p = .788$), whereas RoBERTa significantly outperforms both ($p < .001$).

McNemar’s test was chosen because it compares two classifiers evaluated on the exact same items, rather than on independent samples, unlike a plain accuracy comparison. It looks at the discordant cases and checks whether the disagreements are symmetric or whether one system tends to be favored. Since humans and Qwen were evaluated on the same 117 items, this makes it the perfect test for this case.

It should be noted, however, the absence of a statistically significant performance difference should not be interpreted as evidence that the two systems behave equivalently: Humans and Qwen disagree on 47% of the decided items.

5.2 Linguistic regularities of the corpus

The strongest corpus-level difference concerned **numerical references**. Truthful reviews contained substantially more numerical information than deceptive ones (1.145 vs. 0.292 references per 100 words; $\delta = -0.353$). This pattern was consistent with a broader tendency for truthful reviews to contain more concrete contextual information, in fact temporal references were also more frequent in truthful reviews (1.520 vs. 1.071; $\delta = -0.172$), while spatial references showed the same direction but a considerably smaller effect ($\delta = -0.080$).

Truthful reviews were also characterized by a more balanced affective profile. They contained more negative-affect terms (1.049 vs. 0.633; $\delta = -0.235$) and more contrast markers (0.827 vs. 0.545; $\delta = -0.185$). The latter result is particularly consistent with the human questionnaire, in which the presence of both positive and negative aspects was the cue most associated with truthfulness.

Several features instead showed higher values in deceptive reviews. Promotional language was more frequent (1.403 vs. 1.066; $\delta = +0.178$), as were exclamation marks ($\delta = +0.143$) and positive-affect terms ($\delta = +0.140$). Deceptive reviews also had slightly longer sentences ($\delta = +0.150$), higher readability-complexity scores according to ARI ($\delta = +0.123$), more perceptual terms ($\delta = +0.132$), repetition ($\delta = +0.106$), and certainty markers ($\delta = +0.103$). These effects, however, were generally smaller than those observed for numerical references and negative affect.

A particularly notable result concerned **first-person language**. First-person pronouns were substantially more frequent in deceptive reviews than in truthful ones (6.40 vs. 4.71 per 100 words; $\delta = +0.320$), making this the second-largest effect in absolute magnitude. This direction contrasts with the human participants’ responses, where personal language was more often associated with truthful reviews.

This result shows an important gap between the cues people perceive as diagnostic and the regularities that actually exist at the corpus level.

Not all theoretically motivated features distinguished the two classes. Total review length showed a negligible effect ($\delta = -0.044$, $p_{\text{FDR}} = .297$), as did hedging ($\delta = -0.042$, $p_{\text{FDR}} = .052$) and cognitive-process language ($\delta = +0.034$, $p_{\text{FDR}} = .310$). These features were therefore not considered robust corpus-level differences.

Finally, we examined whether the observed patterns were driven by particular hotels. Cliff’s δ was recomputed separately within each of the five original hotel-based folds. Numerical references, first-person language, negative affect, promotional language, temporal references, mean sentence length, exclamation marks, perceptual language, readability, and spatial references showed the same direction in all five folds. Contrast markers showed the corpus-level direction in four folds, while the fifth fold was effectively neutral ($\delta = +0.009$). Thus, the main linguistic patterns were broadly stable across hotel groups, although this analysis does not remove the broader source confound between truthful TripAdvisor reviews and deceptive MTurk reviews.

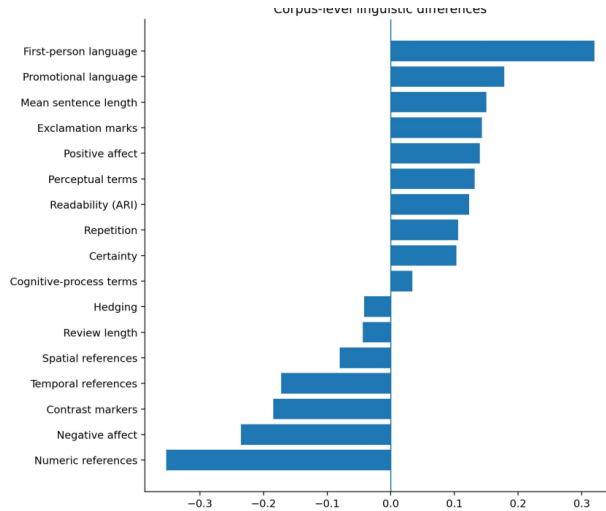


Figure 6: Cliff’s delta of the 17 linguistic proxies.

5.3 What cues are associated with the decisions of Human, Qwen, and RoBERTa?

To compare humans, Qwen, and RoBERTa, we chose not to reduce the analysis to a simple binary classification (truthful/deceptive), as this would result in a significant loss of information: two reviews both classified as deceptive could reflect very different levels of confidence. Therefore, for each system, we adopted a continuous score oriented in the same direction (a higher score $\rightarrow$ a greater bias toward deceptive):

- **Humans:** the Human Deception Rate, the proportion of annotators who judged the review as deceptive (available for 120 reviews);
- **Qwen:** the `qwen_deception_score`, to be interpreted as a decision margin and not as a calibrated probability;
- **RoBERTa:** $p_{\text{deceptive}}$, the probability estimated by the classifier for the deceptive class.

To verify whether linguistic features are associated with the decisions of the three systems, the Spearman correlation was calculated between each of the 17 features and the respective deception score.

This approach, however, can run into a confounding problem: if a given feature is almost always present in reviews of one label, it is hard to tell whether the model is actually using that feature to decide, or whether the feature happens to correlate with the label, without playing any real role in the model’s decision.

To mitigate this problem, a partial Spearman correlation was calculated controlling for the gold label. This isolates the effect of the individual feature, allowing us to answer the question: given two reviews with the same actual label, does a higher value of that feature shift the model’s output toward classifying the review as truthful, or as deceptive?

Specifically, features and scores were transformed into ranks, the component explained by the gold label was removed from both via regression, and the resulting residuals were correlated with each other, obtaining an adjusted ρ coefficient. A $\rho_{\text{adjusted}} > 0$ indicates that, given the same gold label, an increase in the feature is associated with a greater tendency for the model to classify the text as deceptive; a value $\rho_{\text{adjusted}} < 0$ indicates the opposite tendency.

Given the multiplicity of tests (17 features), an FDR correction was finally applied, calculated separately for each system.

For humans, none of the 17 features survives the FDR correction. This result is not necessarily in contradiction with the data obtained from human participants. Indeed, the questionnaire only tells us which cues humans say they used, not which cues actually determined their decisions. This control, instead, makes it possible to verify whether quantitative variations in feature proxies are actually associated with the tendency to respond deceptively. These are two distinct issues, especially since linguistic proxies are simplified by construction, and the human sample is limited to 120 reviews.

For Qwen, 9 of the 17 features survived the FDR correction. The strongest association was with positive affect: given the same gold label, more positive language pushes Qwen toward classifying a review as truthful (opposite direction from the corpus distribution). Temporal references and perceptual language showed similar, but weaker associations. The remaining significant associations were: review length ($\rho = +.116$), cognitive process ($\rho = +.099$), numeric references ($\rho = +.096$), spatial references ($\rho = -.094$), exclamation ($\rho = +.086$), and negative affect ($\rho = +.080$).

Among these, the case of numeric references is particularly notable: in the corpus, this feature is strongly associated with truthful reviews ($\delta = -.353$), while Qwen, with the same gold label, associates it with a greater tendency toward deceptiveness. The same misalignment is observed for negative affect. Overall, of the nine significant associations for Qwen, only five are directionally consistent with the regularities of the corpus.

For RoBERTa, nine associations are also significant after FDR correction, but with a much stronger directional alignment with the corpus. The strongest association again concerns numeric references ($\rho = -.236$, $p_{\text{FDR}} < .001$): with the same gold label, more numeric references shift RoBERTa towards truthfulness, in line with the corpus distribution. Promotional language ($\rho = +.146$, towards deceptive, consistent with the corpus) and review length ($\rho = -.139$, towards truthful) also show significant associations. However, the latter should be treated with caution, as length does not differ significantly between the two gold classes after FDR correction, and should therefore be understood as a characteristic of RoBERTa’s behavior rather than a robust corpus-wide cue.

The overall picture that emerges from this analysis represents one of the key findings of the experiment. RoBERTa, having been fine-tuned on the labels of this specific corpus, was able to directly learn which linguistic patterns in this dataset distinguish truthful from deceptive reviews; consistently, its behavior is much more aligned with the actual regularities of the data (8 out of 9 associations in the direction observed in the corpus). Qwen, however, operates in zero-shot mode and without specific supervision on the 800 reviews in the dataset, relying on general knowledge and heuristics learned during pretraining, which may be generally valid but do not necessarily coincide with the specificities of this corpus. This difference helps explain why RoBERTa achieves an accuracy of approximately 94%, compared to significantly lower results for Qwen.

An important interpretative limitation should be noted, however: RoBERTa’s high alignment with corpus regularities does not demonstrate that the model has learned a general and universal theory of deception. More precisely, it demonstrates that RoBERTa effectively learned the regularities that, in this specific dataset, separate the truthful and deceptive classes. Given the source-label confounding problem discussed previously,

this distinction is crucial: very high accuracy may reflect the learning of genuinely deceptive cues, but it may also reflect, at least in part, the learning of specific regularities related to the provenance and construction of the corpus.

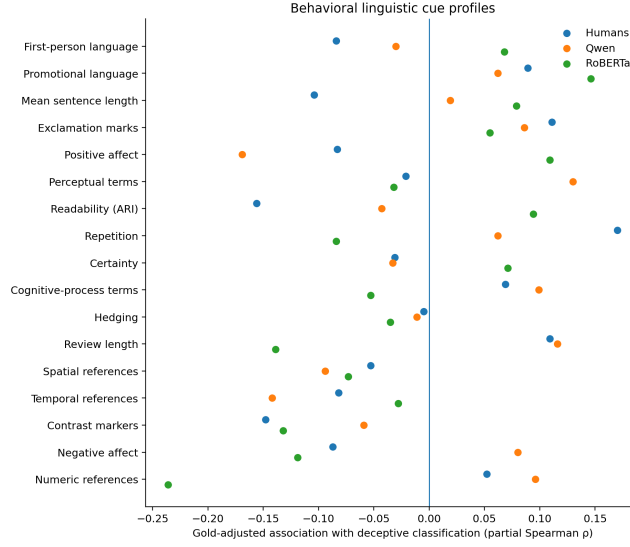


Figure 7: Behavioral profiles of cues.

5.4 Cue conflict

For better contextualization, an additional check, called *cue conflict*, was performed. The intuition behind it is very simple: if a classifier is using particular regularities to produce its predictions, this should become particularly evident when those regularities are violated. Specifically, if a review with a given real label exhibits an unnatural presence of a feature generally correlated with the opposite label, a conflict is generated, and it should be significantly more difficult for the model to classify it correctly.

Cue conflict is therefore a metric that allows us to measure how much the linguistic profile of a sentence points toward the opposite class of its real label.

Only the 14 features that showed significant differences between the classes at the corpus level were analyzed. For each of these, the conflict was calculated separately for each review, and the value of the review was contextualized within its label. This allows the analysis to measure how unusual a review is compared to other reviews sharing the same label, rather than simply reproducing the already known difference between truthful and misleading reviews. The 14 values were then averaged to obtain a single cue-conflict score for each review.

Because the human judgments were collected from multiple participants and not as a single prediction for each review, the cue-conflict score was correlated with the proportion of correct human judgments.

The hypothesis is that if the model effectively exploits those features, misclassified reviews will tend to have a systematically higher cue-conflict score than correctly classified ones.

The results were significantly different between the three systems:

- **Humans:** no significant relationship was observed.
- **Qwen:** misclassified reviews showed only slightly higher conflict than correctly classified ones. Therefore, the corpus-level linguistic regularities captured by our features provide little evidence to explain the model’s errors.
- **RoBERTa:** RoBERTa errors had an average conflict score of 0.573 versus 0.495 for correct predictions, with Cliff’s $\delta = 0.539$ and $p_{\text{FDR}} < .001$.

This result is particularly informative when considered in conjunction with the previous alignment analyses. RoBERTa showed the strongest correspondence with the linguistic regularities of the corpus: its behavioral associations at the feature level were largely oriented in the same direction as the differences observed between truthful and deceptive predictions. The cue conflict analysis suggests that this strong alignment may explain not only its high overall accuracy but also part of its error pattern. RoBERTa performs well when reviews follow the regularities that distinguish the two classes, but becomes substantially more error-prone when a review violates these regularities.

In this sense, the same statistical regularities that appear to constitute a strength of the model may also represent its vulnerability. A truthful review may be difficult for RoBERTa if it contains an unusually strong combination of deceptive-like cues, while a deceptive review may be difficult when its linguistic profile is unusually truthful-like.

Qualitative inspection of the 10 RoBERTa errors present in the human subset supports this interpretation. Among the difficult truthful errors, two of them show strong deceptive-like characteristics, such as perceptual, certainty, and promotional language; among the difficult deceptive ones, one contains contrastive evaluation, temporal detail, and the absence of promotional markers, while another shows numeric and temporal detail much higher than the deceptive median. These cases also demonstrate that “detail” is not a unitary construct: numeric, temporal, spatial, and perceptual information can have different directions.

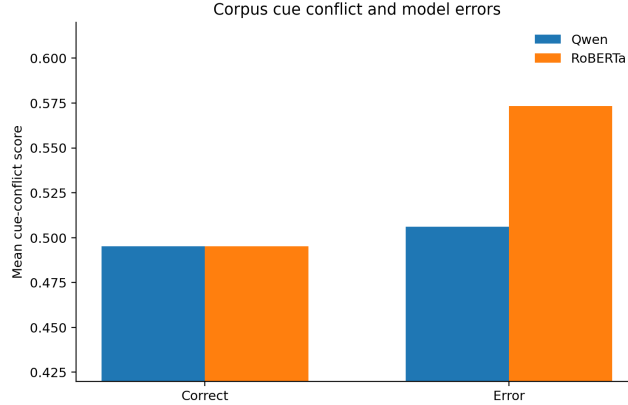


Figure 8: Average cue-conflict for correct and incorrect predictions of Qwen and RoBERTa.

6. Discussion

The main result of this experiment is not so much that RoBERTa surpasses Humans and Qwen in accuracy but rather that the three systems seem to seek and respond to different linguistic regularities.

Humans report using many heuristics, but no single proxy turns out to be consistently associated with their aggregate decisions. This is consistent with the very high heterogeneity observed among participants’ decisions, and suggests either that humans apply highly diversified strategies, or that the strategies they use are, for the most part, difficult to quantify and identify in the text without taking into account other factors, such as personal background, plausibility, and so on.

Qwen shows a selective profile that is partly divergent from the corpus. This is compatible with the fact that, being zero-shot, the model has to rely solely on what it already possesses to classify the two categories (presumably general linguistic knowledge, social knowledge, and prior beliefs about deception). Its behavior deserves further investigation because some cue-association patterns that diverge from the local corpus distribution are nonetheless compatible with linguistic tendencies previously reported in deception research. It would therefore be interesting to see if, for example, Qwen is applying a mix of “plausible stereotypes” about credibility and deception that do not necessarily match the local cue distribution of the corpus.

It’s also important to stress that this is not Qwen’s behavior “in itself,” but rather its behavior under the specific prompt chosen.

Unlike humans and Qwen, RoBERTa has the chance to actually learn the task. By the end of the experiment, RoBERTa emerges as an “ecological learner”: it proves able to learn, with remarkable effectiveness, which linguistic features predict the two classes. Not surprisingly, it is strongly aligned with the corpus regularities, and the cue-conflict score predicts its errors with a substantial effect size.

However, this does not demonstrate that the model has identified and is relying on universal psychological processes of deception. Such high accuracy could plausibly be explained by the model exploiting various shortcuts, especially considering the different collection methods behind the two classes. One clue pointing to this possibility is first-person language: in the literature it is typically associated with truthful reviews, while in this specific corpus it is instead associated with deceptive ones, and RoBERTa’s predictions align with this corpus-specific pattern, not with the deceptive literature. This is interesting for two reasons: first, it highlights that our corpus contradicts some classic cues from the literature; second, it shows that RoBERTa adapts to these contradictions as well.

Moreover, its error profile suggests that this corpus adaptation is not merely correlated with success but may also define a characteristic vulnerability: atypical examples that violate class-typical cue patterns are

disproportionately difficult for the model.

This is extremely interesting from the perspective of deception psychology, because deception is precisely a problem of strategic behavior. A liar does not necessarily produce spontaneous "cues of deception", on the contrary, they may actively try to appear credible. If they know or intuit that an authentic review tends to look balanced, concrete, and not too perfect, they may deliberately introduce, for instance, a small criticism.

This comparison shows that "detecting deception" does not correspond to a single linguistic strategy. Similar or different performance levels can emerge from very different cue profiles, and a responsible NLP analysis must keep separate: what the corpus contains, what humans report, what systems actually do, and what we can genuinely infer about their decision-making mechanism.

7. Limitations

The experiment presents several limitations:

- **Structural confound between source and label:** In the corpus, source and label coincide (truthful reviews from TripAdvisor, deceptive reviews from MTurk). Linguistic differences may therefore reflect deception, TripAdvisor/MTurk writing style, data collection instructions, or other domain-specific characteristics, rather than deception per se.
- **Human sample:** The human sample is a convenience/snowball sample, small in size and not representative of a broader population.
- **Simplicity of the linguistic proxies:** The linguistic proxies used are deliberately simple and transparent, and may not capture more complex or interactive cue patterns.
- **Computational constraints:** Free Google Colab limited what was computationally feasible, ruling out fine-tuning larger models and running multiple seeds/replications.
- **No causal evidence for feature use:** The analysis does not demonstrate that the models are actually relying on the studied cues for their predictions; the associations found are correlational.

8. Future Work

Future work could build on experimental results, mitigating the limitations above and shedding more concrete light on the phenomena addressed here. Promising directions include:

- **Causal validation of RoBERTa’s cues:** Using permutation tests, ablations, and counterfactual text modifications to verify whether RoBERTa actually relies on features such as numerical details, promotional language, and mixed evaluation, rather than merely correlating with them.
- **Generalization test:** Training RoBERTa on this corpus and testing it on an independent, more naturalistic dataset, to assess how much it generalizes beyond the corpus-specific regularities.
- **Human evaluation of model errors:** Having a larger number of human raters evaluate the 50 reviews misclassified by RoBERTa, alongside a set of correctly classified controls, to distinguish errors that are intrinsically difficult from errors that are specific to the model.
- **Cross-model replication:** Replicating the same zero-shot protocol with different LLM families and sizes, to determine whether the biases, prompt sensitivity, and cue associations observed in Qwen generalize across models.
- **More naturalistic deception corpora:** Using corpora involving partial, strategic, and naturalistic deception, rather than only fully fabricated reviews.

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